

Time Series

Applied Time Series Analysis STA457 Lecture Notes

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Time Series Analysis

Forecasting

The Forecasting or Prediction Problem

- 1 In this chapter, we discuss techniques for forecasting the future behaviour of a time series, given present and past values.
- 2 ARMA models are popular tools for obtaining such forecasts. For the most part, we shall assume that the model is known exactly, including specific values for all the parameters. Although this is never true in practice, the use of estimated parameters for large sample sizes does not seriously affect the results.
- 3 The ARMA forecasting techniques are based on the assumption that the transition to the future is guided by its correlation to the past.
- 4 Let us assume the observations $x_1, x_2, \dots, x_t$ are given from a known ARMA model. Then, our goal is to find an optimal predictor for x_{t+l} .
- 5 $\hat{x}_t(l)$ is a predictor of x_{t+l} given data up to x_t .
- 6 The time point t is referred to as the *forecast origin*. In most cases t is the present time.
- 7 The parameter l is the *lead time*, i.e., the number of time units (*steps ahead*) into the future to be forecast.
- 8 The *best* or *optimal predictor* will be in the mean square sense. That is, we will search for a predictor, $\hat{x}_t(l)$, of x_{t+l} that minimizes

$$\mathbb{E} \left[\left(X_{t+l} - \hat{X}_t(l) \right)^2 \right]$$

Definition (Integrable Random Variables)

- ① A random variable is said to be **integrable** if

$$E[|X|] < \infty$$

- ② In other words, integrable random variables have well-defined finite means.

Definition (Square-integrable Random Variables)

- ① A random variable is said to be **square-integrable** if

$$E[|X|^2] < \infty$$

- ② In other words, square-integrable random variables have well-defined finite means as well as finite variances.

Optimal predictor in mean square sense

The *best* or *optimal predictor* will be in the *mean square sense* (MSE). That is, we will search for a predictor, $\hat{x}_t(l)$, of x_{t+l} that minimizes

$$\mathbb{E}[(X_{t+l} - \hat{X}_t(l))^2]$$

Question

Prove that $\mathbb{E}[Y|X = x]$ is closest to Y in the *mean square sense* among all square-integrable random variables that are functions of X . Find the function $h(X)$ of X is optimal predictor of Y .

proof

$$\begin{aligned}
 & \min_{h(x)} E[(Y - h(x))^2] \\
 E[(Y - h(x))^2] &= E\left\{ [Y - E[Y|X]] + (E[Y|X] - h(x)) \right\}^2 \\
 &= E[(Y - E[Y|X])^2] + E[(E[Y|X] - h(x))^2] + \underbrace{2E[(Y - E[Y|X])(E[Y|X] - h(x))]}_0
 \end{aligned}$$

$$\begin{aligned}
 & E_{X,Y} [(Y - E[Y|X]) (E[Y|X] - h(x))] \\
 &= E_X \left[E_{Y|X} [(Y - E[Y|X]) (E[Y|X] - h(x))] \right] \\
 &= E_X \left[\underbrace{E_{Y|X} [Y E[Y|X]] - (E[Y|X])^2 - h(x) E_{Y|X} [Y]}_{=0} + h(x) E[Y|X] \right]
 \end{aligned}$$

proof

$$= E_X \left[\underbrace{(E[Y|X])^2} - \underbrace{(E[Y|X])^2} - \underbrace{h(X)E[Y|X]} + \underbrace{h(X)E[Y|X]} \right] = 0$$

The loss function gets minimized when

$$h(X) = E[Y|X]$$

↑
the conditional expectation of Y given X
is the best predictor

Optimal Linear Predictors

- ① The results we just presented extend to the case in which $X_1, X_2, \dots, X_t$ are random variables such that $\mathbb{E}[X_j^2] < \infty$ for $j = 1, 2, \dots, t$.
- ② In this case, the conditional expectation $\mathbb{E}[X_{t+1}|X_1, \dots, X_t]$ is the *best predictor* of X_{t+1} in the *mean square sense* among all square-integrable functions of the random variables $X_1, X_2, \dots, X_t$.
- ③ Thus, since our goal is to forecast X_{t+1} based on $X_1, X_2, \dots, X_t$, it follows that $\mathbb{E}[X_{t+1}|X_1, \dots, X_t]$ is the *optimal predictor*.
- ④ Although $\mathbb{E}[X_{t+1}|X_1, \dots, X_t]$ is the desired optimal predictor, it is not necessarily *linear* and may be very difficult to obtain in practice.
- ⑤ As a consequence, we will choose to restrict our attention to *linear predictors* and search for *optimal predictors* of the form

$$Z_t = a_1 X_t + a_2 X_{t-1} + \dots + a_t X_1$$

Theorem

If X_t is a stationary time series, then Z_t^* is the best linear predictor of X_{t+1} (has minimum mean square error among linear predictors of the form $Z_t = a_1 X_t + a_2 X_{t-1} + \dots + a_t X_1$) if and only if

$$\mathbb{E}[(X_{t+1} - Z_t^*)X_j] = 0, \quad j = 1, 2, \dots, t.$$

proof

$\Leftarrow$ Let z_t^* be such that $E[(X_{t+l} - z_t^*) X_j] = 0$ for all $j=1, \dots, t$, and let z_t be an arbitrary linear combination of $X_1, X_2, \dots, X_t$. Then,

$$\begin{aligned}
 E[(X_{t+l} - \overset{\downarrow}{z_t})^2] &= E[(X_{t+l} - \overset{\nearrow}{z_t^*} + \overset{\nearrow}{z_t^*} - z_t)^2] \\
 &= E[(X_{t+l} - \overset{\nearrow}{z_t^*})^2] + E[(\overset{\nearrow}{z_t^*} - z_t)^2] + \underbrace{2E[(X_{t+l} - \overset{\nearrow}{z_t^*})(\overset{\nearrow}{z_t^*} - z_t)]}_{0} \\
 &\geq E[(X_{t+l} - \overset{\nearrow}{z_t^*})^2]
 \end{aligned}$$

proof

⇒ To show that $A \Rightarrow B$, we will prove $\neg B \Rightarrow \neg A$

We assume $E[(X_{t+h} - Z_t^*)X_j] = a \neq 0$ for some j .

Roadmap: - show that Z_t^* does not have minimum MSE by producing a linear combination, Z_t , of $X_1, X_2, \dots, X_t$, with smaller MSE than Z_t^* .
 - That is, we will show that $E[(X_{t+h} - Z_t^*)^2] \geq E[(X_{t+h} - Z_t)^2]$

Suppose $E[(X_{t+h} - Z_t^*)X_j] = a \neq 0$ and let $Z_t = Z_t^* + \frac{aX_j}{E[X_j^2]}$

Then,

proof

$$\begin{aligned}
 E[(X_{t+h} - \bar{z}_t)^2] &= E[(X_{t+h} - z_t^* + z_t^* - \bar{z}_t)^2] \\
 &= E[(X_{t+h} - z_t^*)^2] + E[(z_t^* - \bar{z}_t)^2] - 2E[(X_{t+h} - z_t^*)(z_t^* - \bar{z}_t)] \\
 &= E[(X_{t+h} - z_t^*)^2] + E\left[\left(\frac{ax_j}{E[X_j^2]}\right)^2\right] - 2E\left[(X_{t+h} - z_t^*) \frac{ax_j}{E[X_j^2]}\right] \\
 &= E[(X_{t+h} - z_t^*)^2] + \frac{a^2 E[X_j^2]}{(E[X_j^2])^2} - \frac{2a^2}{E[X_j^2]} \\
 &= E[(X_{t+h} - z_t^*)^2] - \frac{a^2}{E[X_j^2]}
 \end{aligned}$$

Since $a \neq 0$, it follows that

$$E[(X_{t+h} - \bar{z}_t)^2] < E[(X_{t+h} - z_t^*)^2]$$

and the result follows.

Forecasting in AR(1) Models

- 1 Consider a centred AR(1) process with a nonzero mean μ given by

$$X_t - \mu = \phi(X_{t-1} - \mu) + W_t$$

- 2 Suppose we want to forecast the process 1 time unit into the future. Note that

$$X_{t+1} - \mu = \phi(X_t - \mu) + W_{t+1}$$

- 3 Taking the conditional expected value (given $X_1, X_2, \dots, X_t$) of both sides, we have:

$$\begin{aligned}\hat{X}_t(1) - \mu &= \phi(\mathbb{E}[X_t | X_1, \dots, X_t] - \mu) + \mathbb{E}[W_{t+1} | X_1, \dots, X_t] \\ &= \phi(X_t - \mu) + \mathbb{E}[W_{t+1}] = \phi(X_t - \mu)\end{aligned}$$

since W_{t+1} is independent of $X_1, \dots, X_t$ and has mean zero.

- 4 So, we have

$$\hat{X}_t(1) = \mu + \phi(X_t - \mu)$$

which means that the forecast for the next value is the process mean, plus some fraction of the current deviation from the process mean.

- 5 If we forecast not just 1 time unit but l time units into the future, we have

$$\hat{X}_t(l) = \mu + \phi(\hat{X}_t(l-1) - \mu), \quad l \geq 1$$

This recursive formula is called the *difference equation* form of the forecasts.

Forecasting in AR(1) Models (Cont'd)

- ① We can solve for a general formula for a forecast with a lead time l in an AR(1) process:

$$\hat{X}_t(l) = \mu + \phi^l(\hat{X}_t - \mu)$$

- ② This relation implies that the fraction of the current deviation from the process mean that is added to μ becomes closer to zero as the lead time gets larger.
- ③ The one-step-ahead forecast error $e_t(1)$ is the difference between the actual value of the process one time unit into the future and the predicted value one time unit ahead.
- ④ For the AR(1) model:

$$e_t(1) = X_{t+1} - \hat{X}_t(1) = [\phi(X_t - \mu) + \mu + W_{t+1}] - [\phi(X_t - \mu) + \mu] = W_{t+1}$$

- ⑤ So, the one-step-ahead forecast error is simply a white noise variable, and it is independent of $X_1, X_2, \dots, X_t$.
- ⑥ Also, the variance of the error would be

$$\text{Var}[e_t(1)] = \sigma_w^2$$

$$\mathcal{I}_t = \{x_t, x_{t-1}, x_{t-2}, \dots\} \cup \{w_t, w_{t-1}, w_{t-2}, \dots\}$$

information set
up to time t

ARMA(2,1): $y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + w_t - \theta_1 w_{t-1}$

$\{y_1, y_2, \dots, y_n\}$

one-step ahead:

$$\hat{y}_{n+1} = \delta + \phi_1 y_n + \phi_2 y_{n-1} + (\hat{w}_{n+1} = 0) - \theta_1 \hat{w}_n$$

two-step ahead: $\hat{y}_{n+2} = \delta + \phi_1 \hat{y}_{n+1} + \phi_2 y_n + (\hat{w}_{n+2} = 0) - \theta_1 (\hat{w}_{n+1} = 0)$

three-step ahead: $\hat{y}_{n+3} = \delta + \phi_1 \hat{y}_{n+2} + \phi_2 \hat{y}_{n+1} + 0 - 0$

AR gradually damp out, MA terms vanish

Rules for forecasting

$$(1) E[X_{t+h} | \mathcal{I}_t] = \hat{x}_{t+h} \quad h > 0$$

$$(2) E[X_{t-h} | \mathcal{I}_t] = x_{t-h} \quad h > 0$$

$$(3) E[W_{t+h} | \mathcal{I}_t] = 0 \quad h > 0$$

$$(4) E[W_{t-j} | \mathcal{I}_t] = w_{t-j} \quad j \geq 0$$

residual at $t-j$

$$= x_{t-j} - \hat{x}_{t-j}$$

Example

$$\text{ARMA}(1,1): Y_t = \delta + \phi Y_{t-1} + W_t - \theta W_{t-1}$$

$$n=200$$

Actual ARMA(1,1)

Residual ARMA(1,1)

198

199

200

1.22

-0.29

$$\hat{y}_{201} = 0.08 + 0.75(1.22) + 0.08(-0.29) = 0.76$$

$$\hat{y}_{202} = 0.08 + 0.75(0.76) + 0.08(0) = 0.65$$

$$\hat{y}_{203} = 0.08 + 0.75(0.65) + 0 = 0.58$$

i

$$\hat{\phi} = 0.75, \hat{\theta} = -0.08,$$

mean $\rightarrow$ intercept = 0.31

$\delta \rightarrow$ constant = 0.08

(δ is not the mean of the series)

$$\underbrace{1.22, 0.76, 0.65, 0.58 \dots}_{10^{12}}$$

$$\text{MA}(2): Y_t = \mu + W_t - \theta_1 W_{t-1} - \theta_2 W_{t-2}$$

$$\begin{aligned} \text{1-step: } Y_{n+1} &= \mu + W_{n+1} - \theta_1 W_n - \theta_2 W_{n-1} & \Rightarrow Y_{n+1} - \hat{Y}_{n+1} &= W_{n+1} \Rightarrow \text{Var}(Y_{n+1} - \hat{Y}_{n+1}) = \sigma_w^2 \\ \hat{Y}_{n+1} &= \mu & & -\theta_1 W_n - \theta_2 W_{n-1} \end{aligned}$$

$$\begin{aligned} \text{2-step: } Y_{n+2} &= \mu + W_{n+2} - \theta_1 W_{n+1} - \theta_2 W_n & \Rightarrow Y_{n+2} - \hat{Y}_{n+2} &= W_{n+2} - \theta_1 W_{n+1} \\ \hat{Y}_{n+2} &= \mu & & -\theta_2 W_n \Rightarrow \text{Var}(Y_{n+2} - \hat{Y}_{n+2}) = \sigma_w^2 (1 + \theta_1^2) \end{aligned}$$

$$\begin{aligned} \text{3-step: } & \\ & \vdots \\ & \vdots \end{aligned} \quad \text{Var}(Y_{n+3} - \hat{Y}_{n+3}) = \sigma_w^2 (1 + \theta_1^2 + \theta_2^2)$$

Forecast Error ($e_t(l)$)

$$e_t(l) = Y_{t+l} - \hat{Y}_t(l) = \mu_{t+l} + W_{t+l} - (\mu_{t+l} + 0) = W_{t+l} \Rightarrow E[e_t(l)] = E[W_{t+l}] = 0$$

* The forecast error is unbiased.

$$\text{Var}(e_t(l)) = \text{Var}[W_{t+l}] = \sigma_w^2$$

it does not depend on the lead time l

ARMA case: (causal and invertible models)

$$Y_{t+1} = \delta + W_{t+1} + \underbrace{\psi_1 W_t + \psi_2 W_{t-1} + \psi_3 W_{t-2} + \dots}_{\text{known}}$$

known

$$\hat{Y}_{t+1} = \delta + \cancel{E[W_{t+1}]} + \psi_1 W_t + \psi_2 W_{t-1} + \psi_3 W_{t-2} + \dots$$

$$Y_{t+1} - \hat{Y}_{t+1} = W_{t+1}$$

$\text{Var}(e_t(1)) = \sigma_w^2$

consider $t+2$

$$Y_{t+2} - \hat{Y}_{t+2} = W_{t+2} + \underbrace{\psi_1 W_{t+1}}_{\text{forecast error}}$$

$\text{Var}(e_t(2)) = \sigma_w^2 + \psi_1^2 \sigma_w^2 = \sigma_w^2 (1 + \psi_1^2)$

conclusion:

$$\text{Var}(e_t(i)) = \sigma_w^2 (1 + \psi_1^2 + \psi_2^2 + \psi_3^2 + \dots)$$

Nonstationary ARIMA:

$$Y_t = \delta + Y_{t-1} + W_t \quad \text{ARIMA}(0,1,0) \quad \text{random walk with drift}$$

$$\hat{Y}_t^{(1)} = \delta + E[Y_t | Y_1, \dots, Y_t] + E[W_t | Y_1, \dots, Y_t] = \delta + Y_t$$

$$\hat{Y}_t^{(l)} = \delta + \hat{Y}_t^{(l-1)} \quad l > 1 \Rightarrow \hat{Y}_t^{(l)} = \delta l + Y_t$$

* The forecast, as a function of the lead time l , is a straight line with slope δ .

$$e_t(l) = W_{t+1} + W_{t+2} + \dots + W_{t+l} \Rightarrow \text{Var}(e_t(l)) = l \sigma_w^2$$

Deterministic Trend Models

$$Y_t = \mu_t + W_t \rightarrow \begin{array}{l} \text{stochastic component} \\ \uparrow \\ \text{deterministic component} \end{array}$$

W_{t+l} is independent of $Y_1, \dots, Y_t$

$$\begin{aligned} \hat{Y}_t(l) &= E[\mu_{t+l} + W_{t+l} | Y_1, \dots, Y_t] = E[\mu_{t+l} | Y_1, \dots, Y_t] + E[W_{t+l} | Y_1, \dots, Y_t] \\ \nearrow \text{forecast at lead time } l & \\ &= E[\mu_{t+l}] + 0 \\ &= \mu_{t+l} \end{aligned}$$

Linear Trend:

$$\mu_t = \beta_0 + \beta_1 t$$

Forecast of response at lead time l :

$$\hat{Y}_t(l) = \beta_0 + \beta_1(t+l) \quad \text{the forecast assumes the same linear trend continues into the future.}$$

Quadratic Trend:

$$Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 \Rightarrow \hat{Y}_t(l) = \beta_0 + \beta_1(t+l) + \beta_2(t+l)^2$$

Periodic Seasonal Trend:

$$\mu_t = \mu_{t+12} \Rightarrow \hat{Y}_t(l) = \mu_{t+l+12} = \hat{Y}_t(l+12) \quad \text{the forecast at lead time } l \text{ is same as the forecast 12 months later}$$

model	Variance of forecast error
nonstationary	$l \uparrow \Rightarrow \text{Var}(e_t(l)) \uparrow$ without bound
stationary	$l \uparrow \Rightarrow \text{Var}(e_t(l)) \uparrow$ but, there is a limit to the increase
deterministic trend	$l \uparrow \Rightarrow \text{Var}(e_t(l))$ constant

ARZMA(1,1,1)

textbook notation for MA

$$\hat{y}_t(1) = \delta + (1+\phi) y_t - \phi y_{t-1} - \theta w_t$$

$$\hat{y}_t(2) = \delta + (1+\phi) \hat{y}_t(1) - \phi y_t$$

⋮

$$\hat{y}_t(l) = \delta + (1+\phi) \hat{y}_t(l-1) - \phi \hat{y}_t(l-2)$$

$$\begin{cases} E[e_t(l)] = 0 & l \geq 1 \\ \text{Var}(e_t(l)) = \sigma_w^2 \sum_{j=0}^{l-1} \psi_j^2 & l-1 \end{cases}$$

nonstationary →

ψ_j weights do not decay to zero
as j increases

⇒ $\text{Var}(e_t(l))$ increases without
bound as l
increases

Exponentially Weighted Moving Average Models (EWMA)

For any invertible model: $Y_t = \pi_1 Y_{t-1} + \pi_2 Y_{t-2} + \dots + W_t \xrightarrow{t+1 \rightarrow t} Y_{t+1} = \pi_1 Y_t + \pi_2 Y_{t-1} + \dots + W_{t+1}$

apply $E[\cdot]$ to
both sides $\Rightarrow \hat{Y}_t^{(1)} = \pi_1 Y_t + \pi_2 Y_{t-1} + \dots$

$$\text{IMA}(1,1): Y_t = Y_{t-1} + W_t - \theta W_{t-1}$$

$$\pi\text{-weights: } \pi_j = (1-\theta)\theta^{j-1} \quad j \geq 1$$

$$\text{one-step ahead forecast} \equiv \text{EWMA} : \hat{Y}_t^{(1)} = (1-\theta)Y_t + (1-\theta)\theta Y_{t-1} + (1-\theta)\theta^2 Y_{t-2} + \dots$$

These weights decay exponentially, and by summing up the geometric series, $\sum = 1$

Recursive Updating Formula: $\hat{Y}_t^{(1)} = \underbrace{(1-\theta)}_{\text{smoothing constant}} Y_t + \theta \hat{Y}_{t-1}^{(1)}$

Prediction Limits

$$y_t = \mu_t + W_t \Rightarrow \hat{y}_t(l) = \mu_{t+l}$$

$$W_t \sim N(0, \sigma_w^2) : e_t(l) = y_{t+l} - \hat{y}_t(l) = W_t \sim N(0, \sigma_w^2)$$

$$\Rightarrow e_t(l) \sim N(0, \sigma_w^2) \quad \left(\text{Var}(e_t(l)) = \sigma_w^2 \right)$$

$$\frac{y_{t+l} - \hat{y}_t(l)}{\sqrt{\text{Var}(e_t(l))}} \sim N(0, 1)$$

So, with probability $(1-\alpha)$, the future observation (l time units ahead), y_{t+l} , falls within

prediction interval $\hat{y}_t(l) \pm Z_{\alpha/2} \sqrt{\text{Var}(e_t(l))}$

this is not confidence interval

In ARIMA models, the forecast error variance depends on $\begin{cases} \sigma_w^2 \\ \psi\text{-weights} \end{cases}$

$$\text{Var}[e_t(l)] = \sigma_w^2 \sum_{j=0}^{l-1} \psi_j^2$$

AR(1):

$$\text{Var}[e_t(l)] = \sigma_w^2 \frac{1 - \phi^{2l}}{1 - \phi^2} \quad \checkmark$$

Example: $\hat{\phi} = 0.5705$, $\hat{\mu} = 74.3293$, $\hat{\sigma}_w^2 = 24.8$

$\hat{y}_t(l) = 70.1479$ $\xrightarrow[\text{prediction interval}]{95\% \text{ CI}}$

$$70.1479 \pm (1.96) \sqrt{(24.8) \frac{1 - 0.5705^2}{1 - 0.5705^2}} = (60.39, 79.91) \quad l=1$$

As l gets larger, for this AR(1) model, both the forecast and prediction limits converge to some fixed long-run values.

$$(60.71, 83.18) \quad l=2$$

$$(62.41, 86.20) \quad l=10$$

$$\begin{aligned}
 \text{MA}(1): \quad & \hat{y}_t(1) = \mu - \theta W_t \\
 & \hat{y}_t(l) = \mu \quad l > 1
 \end{aligned}
 \quad \Rightarrow \quad \text{Var}[e_t(l)] = \begin{cases} \sigma_w^2(1+\theta^2) & l > 1 \\ \sigma_w^2 & l = 1 \end{cases}$$

$(1-\alpha)100\%$ prediction interval:

$$\hat{y}_t(l) \pm Z_{\alpha/2} \sqrt{\text{Var}(e_t(l))}$$

Derive forecasts using ARIMA model

$$\hat{y}_{t+1}(L) = \hat{y}_t(L+1) + \Psi_L \left[y_{t+1} - \hat{y}_t(L) \right]$$

actual forecast error at time $t+1$, which is known once y_{t+1} is observed.

Forecasts are in the original domain of the observations given up to time t .

ARMA

↓
→ \$21K → \$5K → \$5.2K → \$21.1K

If I convert ARIMA into ARMA by differencing, would the forecasts still reside within the domain of the observed data (original domain)?

Differencing → ARMA forecasts $\xrightarrow[\text{sum up the results to obtain forecasts for ARIMA model}]{\text{de-transform (augmentation)}}$

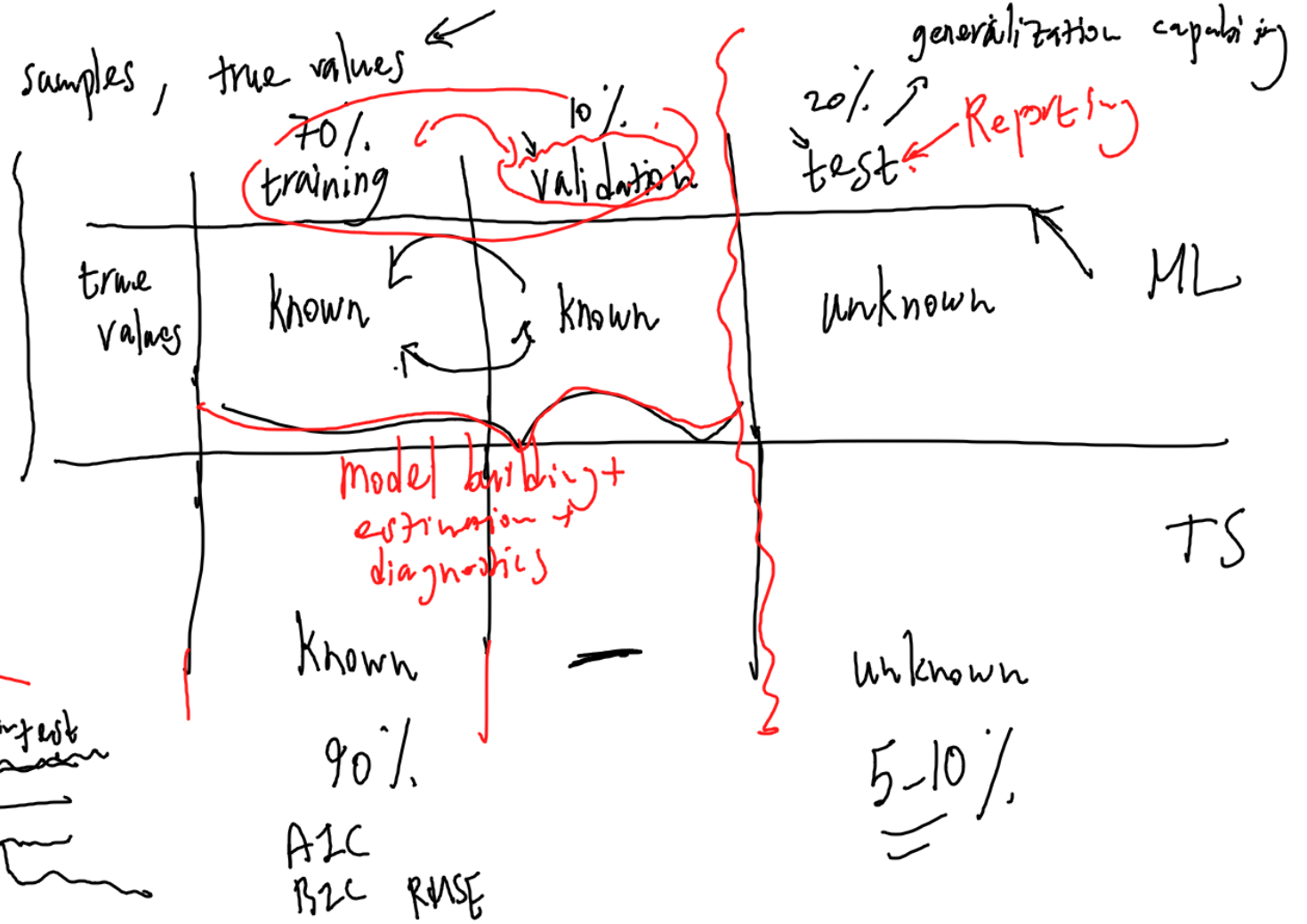
Differencing is a linear operator

$$Z_t = \log(y_t) \Rightarrow \hat{z}_t(L) \rightarrow e^{\hat{z}_t(L)}$$

de-transforming the results $\equiv$ to apply the inverse of transformation

Supervised learning: n samples, true values

AI/ML: model,
parameters fine tuning



order

